

Asian Journal of Communication

ISSN: 0129-2986 (Print) 1742-0911 (Online) Journal homepage: https://www.tandfonline.com/loi/rajc20

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To cite this article: Chenling Shi, Lingling Yu, Nan Wang, Bayi Cheng & Xiongfei Cao (2020) Effects of social media overload on academic performance: a stressor–strain–outcome perspective, Asian Journal of Communication, 30:2, 179-197, DOI: 10.1080/01292986.2020.1748073

To link to this article: https://doi.org/10.1080/01292986.2020.1748073

Published online: 31 Mar 2020.

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ASIAN JOURNAL OF COMMUNICATION
2020, VOL. 30, NO. 2, 179–197
https://doi.org/10.1080/01292986.2020.1748073

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Effects of social media overload on academic performance: a stressor–strain–outcome perspective

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ABSTRACT

As an emerging phenomenon, social media overload has become prevalent among students and led to significant negative consequences. Based on the stressor–strain–outcome model, this study argues that three kinds of overload (information, communication, and social overloads) can influence two psychological strains (technostress and exhaustion) among students and thus affect their behavioral outcome (academic performance). The model is empirically tested through an online survey of 249 Chinese university students who use social media. Results indicate that the three types of social media overload are significant stressors that create technostress, but solely information overload significantly influence exhaustion. In addition, technostress and exhaustion likewise exhibit negative influences on the academic performance of university students.

ARTICLE HISTORY
Received 8 July 2019
Revised 19 March 2020
Accepted 23 March 2020

KEYWORDS
Social media; Overload; Stressor–Strain–Outcome model; Academic performance

The proliferation of Internet has promoted social media to become deeply embedded in people's lives. The China Internet Network Information Center (CNNIC) reports that the number of Internet users in China reached more than 850 million by the end of 2019 with a penetration rate of 61.2%, and these users spend an average of four hours on the Internet per day (CNNIC, 2019). Social media accounts for 32.8% of users' daily Internet time on mobile devices, and WeChat (85.7%) and QQ (68.7%) are the most popular applications (China Internet Watch, 2019). Social media has also been pervasively used in education. Teachers commonly assign homework and initiate discussion via WeChat or QQ groups. The development of social media has enabled university students to search for courses, obtain schedule, and acquire campus information in a timely manner and employ massive learning resources anytime and anywhere. Different mini-programs in social media can effectively promote the combination of online and offline learning modes, thereby greatly facilitating students' learning experience (Tecent Weixiao & New Rank, 2019). Moreover, students can hardly reduce their use of social media applications owing to the integration of social media and mobile payment including WeChat Pay. Therefore, Chinese students tend to develop heavy dependence on social media, which leads to potential negative outcomes.

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Scholars have begun to explore the negative side of social media in the professional context. For example, Brooks (2015) found that social media usage induces technostress and exhibits a negative effect on task performance. Furthermore, van Zoonen, Verhoeven, and Vliegenthart (2016) argued that work-related social media use is a source of boundary conflicts for employees, resulting in emotional exhaustion. Most previous studies on social media are grounded in the business field, but the negative effect of social media on students has received limited attention. The young are a major portion of the social media user population. A report revealed that among Internet users in China, 26% are students, who account for the biggest demographic proportion (CNNIC, 2019). Tecent Weixiao and New Rank (2019) indicated that 80% of university students follow their university's official WeChat account for campus study, and more than half of them check it for more than 30 min per day. As the most active user group, university students are more prone to experience the negative effects of social media than other users. From a pedagogical perspective, understanding the negative side of social media and its associated outcomes is therefore critical and timely.

This study focuses on a particular negative side of social media, which is overload, and propose the following research question: *How does social media overload affect the academic performance of university students?* From a person–environment fit viewpoint, overload comes from an imbalance between environmental demands and a person's ability to cope (Edwards & Cooper, 1990). Pervasive social media access has expanded people's chances to frequently engage in with technology and has forced them to pay continuous attention to the overwhelming volume of social and information demands. Such increased energy requirements may induce overload, which can cause physical and psychological strain (Lee, Son, & Kim, 2016). The social media overload phenomenon has become increasingly evident in daily life and in several academic studies. An individual can be overloaded by social media even with minimal use. He or she does not necessarily have to excessively use social media to experience overload. Problematic social media use (e.g. excessive, compulsive, and addictive use) may also lead to negative effects, but overload represents a prevalent negative side of social media.

Based on the stressor–strain–outcome model, we propose that three types of social media overload (i.e. information, communication, and social overloads) serve as stressors that induce internal strains (i.e. technostress and exhaustion) among students and thus affect their behavioral outcome (i.e. academic performance). This study has several contributions to existing research. First, by focusing on social media overload, we extend research on the negative effect of information technology (IT) to the educational environment. Second, this study uncovers the underlying mechanism of how stressors affect the academic performance of university students, thereby enriching the applicability of the stressor–strain–outcome model. Finally, based on the findings of this study, scholars and practitioners can design interventions or methods to reduce the damage of social media on students.

Theoretical background: stressor–strain–outcome model

The stressor–strain–outcome model provides a framework of stress research and has been applied successfully in previous studies (Cheung & Tang, 2010; Choi, Kim, Lee, & Lee, 2014). Stressors are environmental stimuli that a person encounters and that create

stress (Ayyagari, Grover, & Purvis, 2011). Strain and outcome are the psychological and behavioral reactions of individuals to stressors, respectively (Ragu-Nathan, Tarafdar, Ragu-Nathan, & Tu, 2008; Tarafdar, Cooper, & Stich, 2019). In addition, strain mediates the effect of perceived stressors on outcome variables (Koeske & Koeske, 1993). Previous studies have provided sufficient evidence for understanding stress phenomenon by using the stressor–strain–outcome model in the workplace context. However, the investigation of stressor–strain–outcome model on overload has not been examined sufficiently in the educational context. Combining user-generated content with social networking features, social media can be used for building and maintaining social ties, sharing information, communication, and entertainment. Particularly, social application features, such as pop-up messages and push notifications, multipurpose functions, and self-disclosure features (Salo, Pirkkalainen, & Koskelainen, 2018) can expose university students to excessive information, frequent communication, and social requests. When the demands of usage surpass the processing abilities of students, they may become stressed and feel a sense of losing control over the situations, thereby inducing various kinds of social media overload. These stressful situations further affect students' psychological states of mind and reduce their academic performance.

This study adopts information overload, communication overload, and social overload as stressors in the negative side of social media. Information overload occurs when individuals are exposed to information beyond their capacity to accommodate and process, communication overload indicates the status when communication demands from IT exceed the communication abilities of a person, and social overload describes a situation when people respond to an overwhelming number of social support demands required by their online contacts (Karr-Wisniewski & Lu, 2010; Maier, Laumer, Eckhardt, & Weitzel, 2015). Moreover, we focus on two types of psychological strain, namely, technostress and exhaustion, which are individual psychological states of mind that potentially influence behavioral outcomes through cognitive biases (Steelman & Soror, 2017). Technostress is defined as individuals' psychological states of mind caused by their inability to cope with the current demands induced by their social media usage (Steelman & Soror, 2017). Exhaustion refers to individuals' spontaneous and negative psychological response to stressful stimulus in the environment and represents users' harassed feeling due to social media usage (Maier et al., 2015). From the overload perspective, social media use may create unwanted outcomes. The academic performance of students, which can negatively be affected due to the insufficient time and energy devoted to learning, is a critical behavioral outcome to stressful conditions from the use of social media in the school context. Therefore, the stressor–strain–outcome model is an effective approach to link particular stressors (social media overload), strains (technostress and exhaustion), and outcome (academic performance), thereby emphasizing the negative role of environmental stimuli in affecting the internal state and external behavior of university students.

Research model

Figure 1 illustrates the research model based on the stressor–strain–outcome framework. Previous literature has shown that gender and usage experience can affect students' usage behavior. Thus, these factors are included as control variables in our model.

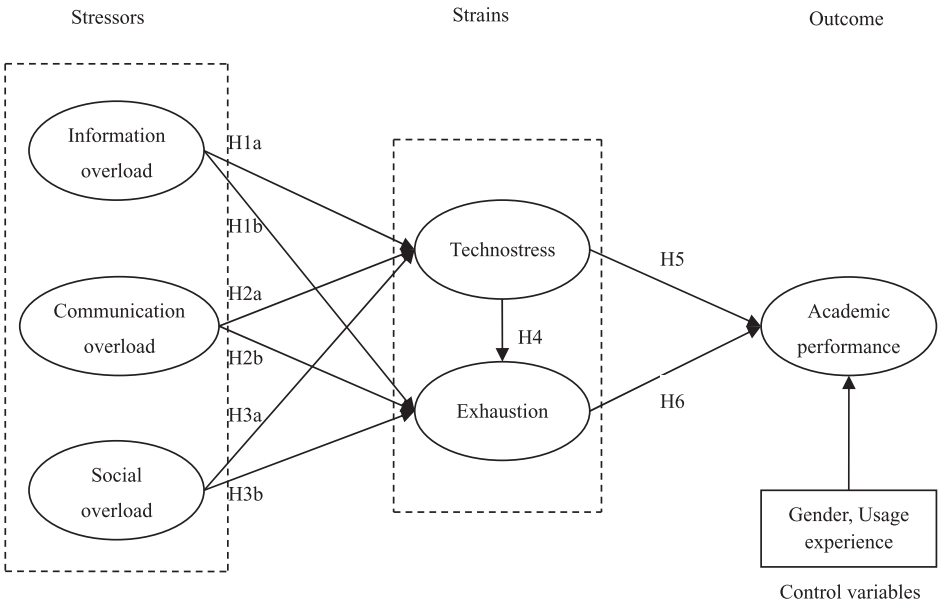


Figure 1. Model with related hypotheses

Overload, technostress, and exhaustion

Information overload

Social media offers university students with various kinds of platforms to obtain extensive amount of information. Students crave for information and news based on their habitual checking behavior. When social media applications push various notifications and renew real-time information newsfeeds and chat discussions, students habitually check the kinds of information and they tend to receive more information than what they would like to receive (Salo et al., 2018). Nevertheless, an individual's capability to deal with information is limited. With an influx of vast information comes an influx of disorder. Students are likely to experience a sense of confusion and suffer from stress when they face overwhelming information (Zhang, Zhao, Lu, & Yang, 2016).Therefore, we propose that:

H1a. Information overload is positively related to technostress.

As social media penetrates into university students' life and learning, they receive, produce, and spread information at an unprecedented rate. Students commonly use social media as a source of information when accomplishing homework, reports, and presentations. Given the abundance of information on social media, individuals may need to spend considerable time and effort to extract effective information (Yu, Cao, Liu, & Wang, 2018). Information overload is likely to occur when students are exposed to information that exceeds their ability to process effectively, which may result in negative emotions and states (Ragu-Nathan et al., 2008). Psycho-cognitive limits can be a serious issue induced by information overload, and students who suffer from cognitive limits tend to feel tired and exhausted. Therefore, we hypothesize the following:

H1b. Information overload is positively related to exhaustion.

Communication overload

The ability to use multiple functions and channels for communication in one device has made social media pervasive in human interactions (Hung, Chen, & Lin, 2015; Wang, Lee, & Hua, 2015). Individuals can be reached and connected anywhere and anytime (Yin, Davison, Bian, Wu, & Liang, 2014). To keep in touch with friends in social media, individuals habitually check their social media applications to avoid missing any messages and respond immediately. All forms of communication messages that are delivered frequently and instantly can lead to cognitive burden among university students and become a source of technostress (Cao & Sun, 2018; Hung et al., 2015). Hence, we propose that:

H2a. Communication overload is positively related to technostress.

Perceived incapacity to deal with excessive communication tasks may trigger exhaustion among social media users (Zhang et al., 2016). People can hardly focus on their current work due to the communication requests on social media and their energy to complete tasks is consumed because of these communication distractions. Research has shown that when workers are interrupted, they need at least 11 min to return to the original work state (Ou & Davison, 2011). For university students, communication requests from numerous social media applications can be disruptive during study hours and thus increase the difficulty to concentrate on their school work and drive them to exhaustion (Lee et al., 2016). Communication overload drives students to exhaustion states by obstructing their primary activities or study tasks. Therefore, we hypothesize that:

H2b. Communication overload is positively related to exhaustion.

Social overload

Self-disclosure and social interaction features of social media help the youth, including university students, who have strong needs to develop their identities on social media and to reveal information about themselves through personal profiles and posts. These social media features allow them to interact and consequently build and maintain relationships with others through the comments under their posts (Salo et al., 2018). University students are at a critical stage in relationship maintenance (Choi & Lim, 2016), and they tend to feel responsible for satisfying the requests embedded in social media. University students are likely to spend a considerable amount of time and energy when using social media due to various unnecessary social connections, activities, and requests embedded in a large social network. If users are socially overloaded in a workplace, they may perceive certain negative experiences of using social media, such as job burnout (Gaudioso, Turel, & Galimberti, 2017) and social media fatigue (Dhir, Yossatorn, Kaur, & Chen, 2018). Similarly, these negative experiences perceived by university students, combined with excessive social support demands, can lead them to a stressful situation, which induces technostress. Accordingly, we hypothesize that:

H3a. Social overload is positively related to technostress.

Social media expansion has allowed university students to interact with various people without the boundaries of time and location. However, exposure to these social interactions exhausts them when they feel uncomfortable in responding to numerous social

requests. Maier et al. (2015) explained that social overload reflects people's feeling of providing overwhelming social support to others, which can cause the exhausted feeling of excessive social connections. Therefore, failing to respond to the overwhelming number of social support requests can be a burden for students and consequently lead to their exhaustion. Therefore, we propose the following hypothesis:

H3b. Social overload is positively related to exhaustion.

Technostress, exhaustion, and academic performance

Exhaustion reflects people's aversive and potentially harmful psychological reaction to stressful situations and characterizes their feeling of tiredness from performing IT activities (Maier et al., 2015). Technostress, which is derived from the use of social media, can make individuals feel exhausted (Yin et al., 2014). Therefore, one critical result of technostress is the feeling of exhaustion. We theorize that technostress induces exhaustion, and this concept accounts for why a growing number of students feel exhausted when using social media. Hence, we hypothesize that:

H4. Technostress is positively related to exhaustion.

Technostress requires additional energy from students as they use social media, and it potentially takes away time from their learning tasks that are necessary for achieving good academic performance. Students with technostress are likely to exhibit an inability to appropriately respond to their learning requirements, because they can hardly balance between technology use and study. Therefore, we infer that:

H5. Technostress is negatively related to academic performance.

Owing to increasing social media penetration, a growing number of people perceive that they face an excessive amount of information and social requests. Students tend to spend a substantial amount of energy on social media and thus become exhausted. In the organizational behavior field, work exhaustion has been found to negatively impact employees' productivity, organizational commitment, and job satisfaction (Swider & Zimmerman, 2010). In the educational context, we infer that exhaustion impairs students' academic performance because they cannot concentrate their energies on learning. Therefore, we hypothesize that:

H6. Exhaustion is negatively related to academic performance.

Research method

Sample and data collection

This study investigated the effects of social media overload in an educational setting. Therefore, the target population was composed of university students who use social media. According to the report launched by CNNIC (2017; 2019), people under the age of 30 accounted for the highest proportion among social media users and most of them are students. In addition, 81% of users have used social media for more than three years. Therefore, student group can be viewed as a representative part of the social

media user population (Cao & Sun, 2018). Given that this study attempts to understand the general effect of overload on student users, social media platforms were not distinguished from one another. Thus, the term 'social media' in this study is a general concept that covers most of the popular and frequently-used social media applications in China (e.g. WeChat, QQ, and Sina Weibo), rather than a specific social media platform or website.

We collected the data by publishing advertisements on five comprehensive Chinese universities' bulletin board systems (BBSs). A BBS is a virtual community where students can post information and comment or communicate with one another. We asked students who intend to participate in the research to comment their email addresses on the advertisement thread. Finally, 573 students were invited to participate in an online survey. Survey invitations were sent to participants via an email message that contains the link to the online questionnaire. The email started with a note informing that the survey is completely anonymous and all the data will solely be used for academic research. A second email was sent after two weeks to encourage participation and facilitate data collection. Since a monetary reward has been proven beneficial in encouraging participation and valid responses (Buhrmester, Kwang, & Gosling, 2011; Mason & Suri, 2012), we sent red packets each containing 10 yuan (approximately 1.52 dollars) to participants' WeChat accounts when they have finished the online questionnaire. Finally, 253 undergraduate students completed the survey, which indicates a response rate of 44.15%. Only those who self-reported as using social media were valid for this study. After eliminating four participants who did not use social media, the remaining 249 respondents were considered valid samples for investigation.

Appendix A lists the demographic characteristics of respondents. They were undergraduates from different grades and were 20 years old on average. The majority of respondents spent over two hours using social media per day and have used social media for more than four years, Therefore, most of these young adults were active and experienced users. This finding suggested that excessive use is likely to occur among these respondents, which supports the appropriateness of our selected sample.

Measurement

Our study adopted five-point Likert scales to measure all the variables. Existing literature served as the basis for measurement development since they are well-established and highly reliable (Cho, Ramgolam, Schaefer, & Sandlin, 2011; Karr-Wisniewski & Lu, 2010; Maier et al., 2015; Ragu-Nathan et al., 2008; Yu, Tian, Vogel, & Kwok, 2010). Items for measuring information overload, communication overload and social overload were adopted from Karr-Wisniewski and Lu (2010), Cho et al. (2011), and Maier et al. (2015), respectively. Technostress was evaluated using four items adapted from Ragu-Nathan et al. (2008), which was modified to fit the context of social media use, whereas exhaustion included three items from the recent study of Maier et al. (2015). Given that GPA calculation standards and methods vary among different universities in China, using GPA to measure academic performance can be difficult. Therefore, we used the items developed by Yu et al. (2010) to capture university students' self-reported performance. Furthermore, gender and usage experience (years of using social media) were selected as control variables to observe if there is any influence of personal characteristics.

Considering that this study investigates Chinese university students, a back-translation approach was used to help minimize misinterpretations in item translation. During the first round of translation, the original items adapted from English literature were translated into Chinese. Thereafter, the Chinese version of the items was translated back into English by another independent translator without referring to the original piece. We appropriately modified several wordings after the two English versions were compared. Moreover, prior to the formal testing, we conducted a preliminary test involving 50 undergraduates who use various kinds of social media applications daily. The results showed satisfactory scales' reliability and validity.

Data analysis

Partial Least Squares (PLS) was adopted to evaluate the proposed model for certain reasons. First, as a second-generation structural equation modeling (SEM) technique, PLS can simultaneously estimate the measurement model and structural model (Chin, Marcolin, & Newsted, 2003). Second, compared with covariance-based SEM, PLS can assess latent variables with non-normal statistical distribution (Bassellier & Benbasat, 2004; Wynne, 1998). Owing to the potentially skewed nature of negative perceptions including technostress (Turel, Serenko, & Giles, 2011), PLS is superior to covariance-based SEM tools. Third, compared with other techniques, PLS is more suitable for testing relatively new relationships and theory exploration (Gefen, Rigdon, & Straub, 2011), which fits the explorative nature of our study. Measurement model and structural model were likewise assessed following a two-step procedure.

Measurement model

Reliability

Reliability was assessed by comparing the values of Cronbach's alpha and composite reliability (CR). Table 1 shows that the values of Cronbach's alpha of all constructs range from 0.72–0.90, and the values of CR exceed 0.80, which meet the threshold value of 0.70 (Fornell & Larcker, 1981; Nunnally, 1994). Therefore, the measurements' adequate reliability is confirmed.

Convergent validity

Fornell and Larcker (1981) and Hair Jr, Hult, Ringle, and Sarstedt (2016) suggested that convergent validity is considered acceptable when (1) item loadings exceed 0.70, (2) average variance extracted (AVE) values exceed 0.50, and (3) CR values are equal to or above 0.70. Table 1 reveals that apart from that item IO1 that loaded lower than 0.70 (above 0.60), other items' loadings are greater than 0.70. The minimum value of AVE is 0.63, which indicates that AVE values meet the recommended level. Furthermore, all CR values are higher than 0.70. Thus, convergent validity was established through these three criteria.

Discriminant validity

The square roots of AVE and correlations were adopted to test discriminant validity (Fornell & Larcker, 1981). Table 2 shows that each construct's square root of AVE

Table 1. Psychometric properties of instruments (N = 249).

Construct	Item	Mean	Std. Dev	Loading	Cronbach's alpha	CR	AVE
Information overload (IO)	IO1	3.67	1.03	0.69	0.72	0.83	0.63
	IO2	2.64	1.06	0.88			
	IO3	3.12	1.03	0.80			
Communication overload (CO)	CO1	3.48	1.14	0.70	0.75	0.85	0.66
	CO2	2.80	1.09	0.91			
	CO3	2.95	1.08	0.82			
Social overload (SO)	SO1	2.74	1.04	0.81	0.88	0.91	0.68
	SO2	3.00	1.06	0.80			
	SO3	2.63	1.07	0.81			
	SO4	2.83	1.10	0.85			
	SO5	3.02	1.12	0.84			
Technostress (TS)	TS1	2.49	1.04	0.84	0.86	0.90	0.70
	TS2	2.76	1.11	0.87			
	TS3	2.80	1.16	0.84			
	TS4	2.87	1.11	0.80			
Exhaustion (EX)	EX1	2.74	1.08	0.90	0.88	0.93	0.81
	EX2	2.81	1.00	0.90			
	EX3	2.63	1.02	0.89			
Academic performance(AP)	AP1	3.59	0.98	0.86	0.90	0.93	0.77
	AP2	3.50	0.97	0.89			
	AP3	3.55	0.98	0.88			
	AP4	3.64	1.00	0.88			

displayed in the diagonal is higher than the correlations in the row and column that correspond to that construct. In addition, Table 3 depicts that all item loadings are significantly higher than their loadings on other constructs, thereby providing evidence for discriminant validity.

Considering that the data were self-reported answers from the same respondent, we adopted several approaches to measure the potential threat of common method bias. First, the result of Harman's single-factor test (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003) showed that no single factor explains most of the variance, in which the most significant factor explained only 34.5% of the variance. Therefore, common method bias does not exhibit a serious influence on the results of this study (Sanchez, Korbin, & Viscarra, 1995). Second, drawing upon the method proposed by Liang, Saraf, Hu, and Xue (2007), we added in the PLS model a common method factor whose indicators included all observed items. Results indicated that the substantive factors explained 71.7% of the overall variance, whereas the method factor accounted for only 0.9% of the variance. Appendix C presents the test results. Moreover, a correlation matrix was presented as a complementary evidence for common method bias assessment. No extremely high correlation ($r > 0.90$) (Pavlou, Liang, & Xue, 2007) was observed in Table 2. Hence, the threat of common method bias can be excluded.

Table 2. Correlations and square roots of AVE (N = 249).

Construct	Mean (SD)	IO	CO	SO	TS	EX	AP
IO	3.14 (0.83)	**0.79**					
CO	3.08 (0.90)	0.61	**0.81**				
SO	2.84 (0.89)	0.31	0.32	**0.82**			
TS	2.73 (0.92)	0.43	0.47	0.41	**0.84**		
EX	2.73 (0.93)	0.44	0.39	0.32	0.65	**0.90**	
AP	3.57 (0.86)	−0.22	−0.16	−0.24	−0.36	−0.35	**0.88**

Note: The square roots of AVE are displayed in the diagonal and in bold text.

Table 3. Cross-loadings of items (N = 249).

Item	IO	CO	SO	TS	EX	AP
IO1	**0.69**	0.42	0.16	0.21	0.23	−0.17
IO2	**0.88**	0.56	0.33	0.45	0.52	−0.23
IO3	**0.80**	0.49	0.25	0.38	0.30	−0.13
CO1	0.43	**0.70**	0.20	0.25	0.18	−0.07
CO2	0.59	**0.91**	0.24	0.54	0.45	−0.21
CO3	0.49	**0.82**	0.36	0.37	0.32	−0.12
SO1	0.29	0.25	**0.81**	0.29	0.30	−0.19
SO2	0.24	0.26	**0.80**	0.31	0.21	−0.19
SO3	0.30	0.30	**0.81**	0.31	0.31	−0.20
SO4	0.28	0.24	**0.85**	0.36	0.24	−0.22
SO5	0.26	0.28	**0.84**	0.42	0.26	−0.18
TS1	0.44	0.45	0.38	**0.84**	0.60	−0.32
TS2	0.37	0.44	0.30	**0.87**	0.52	−0.29
TS3	0.35	0.39	0.37	**0.84**	0.50	−0.31
TS4	0.39	0.43	0.33	**0.80**	0.55	−0.27
EX1	0.45	0.43	0.34	0.59	**0.90**	−0.30
EX2	0.40	0.37	0.26	0.58	**0.90**	−0.31
EX3	0.43	0.33	0.27	0.58	**0.89**	−0.33
AP1	−0.12	−0.13	−0.21	−0.30	−0.26	**0.86**
AP2	−0.22	−0.18	−0.21	−0.30	−0.31	**0.89**
AP3	−0.27	−0.22	−0.17	−0.33	−0.35	**0.88**
AP4	−0.19	−0.12	−0.25	−0.33	−0.31	**0.88**

Structural model

Figure 2 presents the results of the PLS test of the structural model. Most hypothesis are significant, thereby revealing the essential impact of social media-induced overload on university students' academic performance. Information overload exhibits a direct positive relationship with technostress (β=0.19, $p < 0.001$) and social media exhaustion (β=0.21,

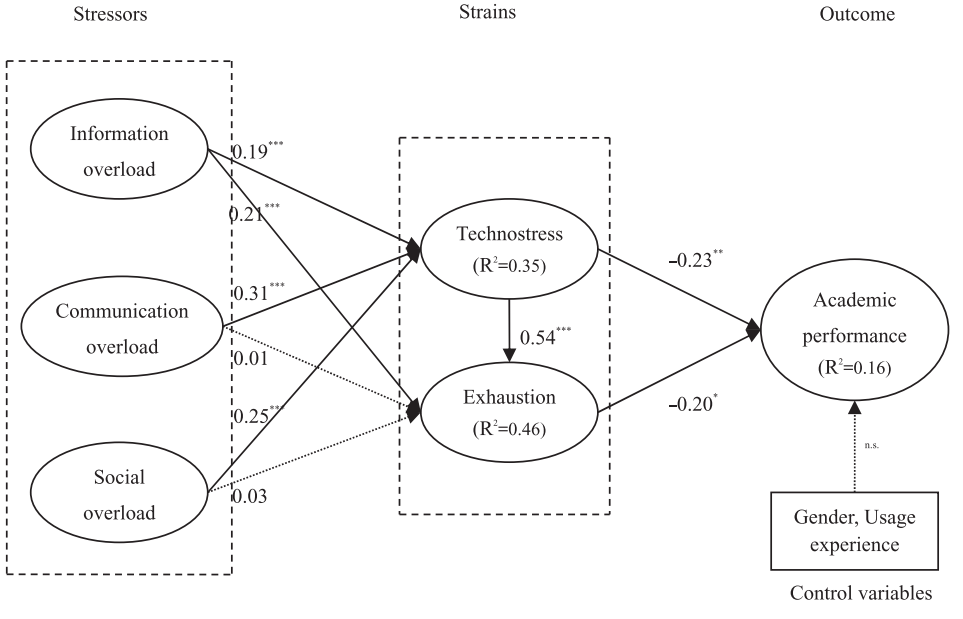


Figure 2. PLS results of the structural model. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

$p < 0.001$). Communication overload and social overload positively affect technostress, with path coefficients of 0.31 and 0.25 ($p < 0.001$), respectively. Nevertheless, their relationships with exhaustion are not supported. The path coefficient of technostress to exhaustion is 0.54 ($p < 0.001$), which indicates a strong connection between the two constructs. In the strain–outcome paradigm, technostress and exhaustion lead to decreased academic performance among university student users, with path coefficients of -0.23 ($p < 0.01$) and -0.20 ($p < 0.05$), respectively. For the control variables, gender and usage experience show no significant effect on the dependent variable. Finally, 16% variance of academic performance is explained.

Discussion and conclusions

Key findings

By applying the stressor–strain–outcome paradigm, this study presents the conceptual framework and empirical evidence for the phenomenon of social media overload and examines its connection with the academic performance of university students. An excessive amount of information, frequent communication, and numerous social support requests can influence the academic performance of university students through their technostress levels and emotional exhaustion. Our results reveal important findings.

First, all three dimensions of social media-induced overload (i.e. information overload, communication overload, and social overload) are significant determinants of technostress. This finding supplements the research of Lee, Chang, Lin, and Cheng (2014), which suggested that overdependence on smartphones is related to compulsive use and can cause technostress. Therefore, overloads from excessive social media use are important stressors that influence university students' technostress. The sources of social media-induced technostress involve informational, communicational, and social aspects. Furthermore, communication overload exerts the highest impact on technostress, followed by social overload and information overload. This finding is inconsistent with the finding of Lee et al. (2016), which indicates that information, communication, and system feature overloads generate almost equal impacts on strain. Our finding may be explained by the fact that frequent communication requirements exert great distraction that induces technostress (Brooks, Longstreet, & Califf, 2017). The result advances our understanding of the effects of overload due to excessive social media use on university students' technostress.

Second, in contrast to the result that information overload has a significantly positive impact on exhaustion, the influences of communication overload and social overload on exhaustion are not significant as expected. This finding is inconsistent with previous studies on social networking sites (SNSs) (Lee et al., 2016; Maier et al., 2015; Zhang et al., 2016) that information, communication, and social overloads have significant effects on SNS fatigue and exhaustion. This phenomenon may occur when students choose to reduce the daily communication and social requests owing to the autonomy in social media use, such as filtering unnecessary requests, or sending posts or information to several targeted friends. Therefore, students with richer usage experience in processing communication and social requests may have less exhaustion risks than those with less usage experience. The finding also reveals that communication and social support

response require less cognitive resources than information process. Students can choose to ignore or delay social and communication requests when they feel that these requests may consume a considerable amount of their energy. Thus, the interactive feature of social and communication requests allows students to save mental resources on unwanted requests (although they can still receive these requests), thus alleviating the level of exhaustion.

Third, technostress and exhaustion can negatively influence university students' academic performance. This finding is in line with the extant research (Brooks & Califf, 2017; Choi et al., 2014), which indicates that technostress and exhaustion are vital factors in influencing performance in an organizational environment. When technostress and exhaustion occur, students may not be able to concentrate on their school work. Thus, they should regulate their behaviors to avoid negative psychological status.

Theoretical implications

The theoretical implications of this study are presented as follows. First, unlike the previous studies that concentrate on the positive effect of social media in an academic setting (Ainin, Naqshbandi, Moghavvemi, & Jaafar, 2015), the theoretical framework proposed in our study uncovers the unexpected negative influence of the use of social media that was initially released to facilitate communication and productivity. Improved Internet connection and social media accessibility have allowed students to easily engage in social media activities during and after their classes. Despite the ongoing debate on the effects of social media in education, heavy exposure to social media has already resulted in negative consequences, such as overload, technostress, and exhaustion, which must be systematically investigated to obtain balanced perspectives of social media (Lee et al., 2016). Our results provide admonitory information about the effects of social media-induced overload on the academic performance of students, thereby improving the understanding of the negative side of social media use.

Second, our study reveals the underlying mechanism of how social media overload affects students' academic performance. The stressor–strain–outcome model has been applied to examine job-related stress (Choi et al., 2014; Choi & Lim, 2016), but it has not been tested in explaining social media-induced stress among university students. In the educational context, a theory-guided empirical research on social media overload is relatively scarce, and the exact mechanism of its impact has not been explored (Alwagait, Shahzad, & Alim, 2015; Junco & Cotten, 2011; Lau, 2017). Our research reveals a clear interplay among environmental stimuli, psychological strains, and behavioral outcomes, which helps to understand the underlying mechanism of how social media overload creates negative effects on student users in a coherent and comprehensive manner.

Third, this study advances our understanding of their irrational use of social media by identifying a comprehensive classification of social media-induced stressors in terms of perceived overload associated with informational, communicational, and social aspects of social media. This study generalizes the research on overload to the educational context, and emphasizes the role of social overload as a social media-specific stressor in creating the adverse effects on psychological reaction (Maier et al., 2015). In addition, this research is conducted in China, which has typical characteristics including collectivism and guanxi culture that influence students' social media use behavior. This study also enriches social media research in the educational environment that has been

investigated in other countries, such as Kuwait (Aladwani & Almarzouq, 2016), Brazil (Felisoni & Godoi, 2018), and Italy (Giunchiglia, Zeni, Gobbi, Bignotti, & Bison, 2018).

Practical implications

This study provides several practical implications. First, although the current university students extensively use social media, the negative consequences of excessive use of social media should not be ignored. By identifying the role of overload, students can recognize their problematic use behaviors to alleviate the negative consequences brought by social media use. For example, students can change the visibility settings on friends' posts, cancel push notifications, or quit unnecessary online social networks to reduce unexpected online social activities. They can also reduce time and energy on social media through delayed response and centralized reply.

Second, by identifying the negative effect of social media overload, university administrators must rethink whether social media should be used as an educational tool or explore on how to effectively utilize social media to minimize its negative effects on students. Despite its substantial benefits, using social media may harm students' learning when it becomes excessive, especially for students with low self-regulation (Rouis, Limayem, & Salehi-Sangari, 2011). However, this finding does not mean that social media should be completely abandoned in educational activities. To help reduce the level of overload, educational practitioners and policy makers can consider the design of social media use when developing their teaching strategies, including time, types of applications, and feedback. In addition, educational institutions should provide coping strategies to regulate students' social media use behaviors, such as offering stress-management countermeasures to help students identify stressful situations (Jena, 2015). They can also educate students to enhance their IT mindfulness, which can effectively alleviate technostress conditions and improve academic performance (Ioannou & Papazafeiropoulou, 2017).

Limitations and future research

Several limitations of this study should be acknowledged. First, the result of our study is based on data collected from universities in China. However, Chinese students and students from Western countries may have diverse interests on social media content and may have developed their own way of use, which leads to differences in usage habits and behaviors. Thus, students from different cultures may perceive different levels of the three types of overload. This perception may affect the identification of the antecedents of technostress and exhaustion. Future research that considers cultural differences are needed to test whether our results can be applied in different regions.

Second, individual variables, such as personality traits, anxiety, and internet self-efficacy, may have direct or indirect effects on students' preferences of social media use and their perception of overload. Therefore, to address this limitation, we encourage future researchers to include these individual differences into a research model and present a comprehensive research on the effect of social media overload on students' academic performance.

Third, cross-sectional data are collected to investigate students' social media usage behavior. However, this approach is limited in investigating the causality of the model

as usage behavior constantly interacts with environmental and cognitive elements (Lee, Kang, & McKnight, 2007). For instance, low academic performance can further induce students to spend extra time in using social media for comfort or to escape from their negative feelings. Therefore, a longitudinal research design is recommended in future research.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the China Scholarship Council: [Grant Number 201706340065], State-owned Assets Collaborative Innovation Center "Research on innovation management of state-owned enterprises": [Grant Number BTBUGZGL201902], and National Natural Science Foundation of China: [Grant Number 71602006].

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Appendices

Appendix A. Demographic information of respondents

Respondent (N = 249)	Mean	Min-Max
Age	20.4	16–25
	Frequency	Percentage
Gender		
Male	109	43.8%
Female	140	56.2%
Grade		
Freshman	42	16.9%
Sophomore	61	24.5%
Junior	67	26.9%
Senior	79	31.7%
Social media usage	Frequency	Percentage
Duration of everyday use		
<1 h	11	4.4%
1–2 h	64	25.7%
2–3 h	60	24.1%
>3 h	114	45.8%
Frequency of everyday use		
1–5 times	34	13.6%
6–10 times	58	23.3%
11–15 times	54	21.7%
>15 times	103	41.4%
Use experience		
<1 year	1	0.4%
1–3 years	29	11.6%
4–6 years	110	44.2%
7–9 years	71	28.5%
>9 years	38	15.3%

Appendix B. measures of constructs

Information Overload	
IO1	I am often distracted by the excessive amount of information in social media.
IO2	I am overwhelmed by the amount of information that I process on a daily basis from social media.
IO3	I feel some problems with too much information in social media to synthesize instead of not having enough information.
Communication Overload	
CO1	I feel that I generally receive too many notifications on new postings, push messages, and news feeds, among others from social media as I perform other tasks.
CO2	I often feel overloaded with social media communication.
CO3	I receive more communication messages and news from friends on social media than I can process.
Social Overload	
SO1	I take too much care of the well-being of my friends on social media.
SO2	I deal with my friends' problems on social media too much.
SO3	My sense of responsibility for how much fun my friends have on social media is too strong.
SO4	I care for my friends on social media too often.
SO5	I pay too much attention to my friends' posts on social media.
Technostress	
TS1	I am forced by social media to live very tight time schedules.
TS2	I am forced to change habits to adapt to new developments on social media.
TS3	I have to sacrifice my personal time to keep current on new social media updates.
TS4	I feel my personal life is being invaded by social media.

(*Continued*)

Continued.

Information Overload	
Exhaustion	
EX1	I feel drained from activities that require me to use social media.
EX2	I feel tired from my social media activities.
EX3	Using social media is a strain for me.
Academic Performance	
AP1	I am confident about the adequacy of my academic skills and abilities.
AP2	I feel competent conducting my course assignments.
AP3	I have learned how to successfully perform my coursework in an efficient manner.
AP4	I have performed academically as well as I anticipated I would.

Appendix C. test of common method bias

Construct	Indicator	Substantive Factor Loading (R1)	$R1^2$	MethodFactor Loading(R2)	$R2^2$
Information overload (IO)	IO1	0.906***	0.821	−0.198***	0.039
	IO2	0.666***	0.444	0.216***	0.047
	IO3	0.845***	0.714	−0.046	0.002
Communication overload (CO)	CO1	0.890***	0.792	−0.191***	0.036
	CO2	0.779***	0.607	0.136**	0.018
	CO3	0.806***	0.650	0.020	0.000
Social overload (SO)	SO1	0.826***	0.682	−0.015	0.000
	SO2	0.841***	0.707	−0.044	0.002
	SO3	0.786***	0.618	0.036	0.001
	SO4	0.857***	0.734	−0.011	0.000
	SO5	0.805***	0.648	0.034	0.001
Technostress (TS)	TS1	0.697***	0.486	0.155*	0.024
	TS2	0.980***	0.960	−0.129*	0.017
	TS3	0.897***	0.805	−0.060	0.004
	TS4	0.768***	0.590	0.037	0.001
Exhaustion (EX)	EX1	0.854***	0.729	0.064	0.004
	EX2	0.938***	0.880	−0.045	0.002
	EX3	0.902***	0.814	−0.020	0.000
Academic performance (AP)	AP1	0.895***	0.801	0.047	0.002
	AP2	0.902***	0.814	0.008	0.000
	AP3	0.847***	0.717	−0.044	0.002
	AP4	0.876***	0.767	−0.011	0.000
Average		**0.844**	**0.717**	**−0.003**	**0.009**